

From Powder to Comparable Transport Data: A Faceted Review of LLZO Synthesis, Phase Control, and Densification

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Abstract

Garnet-type $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$ (LLZO) is discussed through processing routes, substitutions, firing schedules, environmental controls, and transport measurements that are often reported in incompatible combinations. This review maps 46 audited bibliographic records into an evidence-limited process chain: powder synthesis, phase/composition control, thermal treatment and densification, and structure/transport readout. Composition, lithium inventory, additives, and process environment are treated as cross-cutting modifiers rather than serial stages. A separate evidence facet defines the minimum record needed to classify two results as directly comparable, parallel qualitative evidence, or insufficiently specified. The collection is standard-scale, not comprehensive: 25 records include locally stored abstracts, 21 are title-only, and full-text checks were selective. Accordingly, the review reports abstract-level associations and topic coverage without inventing unobserved conditions or universal route rankings. Six research gaps are framed as limitations of discrimination in this collection and paired with falsifiable next actions. A boxed precursor-ranking diagnostic reports the actual model Top-5 only as ranked sets, preserves entity-normalization warnings, and explicitly remains unverified and not for laboratory use. The result is an auditable map of what was varied, what was measured, and what must be recorded before LLZO processing claims can be compared.

1 Introduction

LLZO research links a nominal garnet composition to several distinct questions: how a precursor mixture becomes a powder, how composition and substitution relate to phase state, how a powder becomes a ceramic, and how the resulting specimen is measured. A broad review in the audited set likewise spans crystal structure, preparation, doping, and conductivity rather than treating synthesis as an isolated operation [24]. The difficulty is therefore not a shortage of processing labels. It is that a label such as “solid state,” “sol-gel,” or “spark plasma sintering” rarely specifies the complete chain that determines what was ultimately compared.

The stored abstracts illustrate this coupling without licensing a universal mechanism. A solid-state processing study reports sensitivity to lithium inventory and powder geometry along its route to Al-containing LLZO [8]. A solution-based study focuses on precursor homogeneity and follows compositional and structural evolution [22]. A densification study brings powder-size distribution, heating schedule, support geometry, spatial uniformity, and transport into the same reported case [6]. These examples make a route-only catalogue inadequate: upstream composition, firing boundary conditions, specimen geometry, and measurement definitions can all change at once.

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This review therefore contributes an A+B+C bundle. Contribution A is a faceted process framework whose primary chain is *powder synthesis* $\rightarrow$ *phase/composition control* $\rightarrow$ *thermal treatment/densification* $\rightarrow$ *structure/transport readout*. Composition and environment variables remain cross-cutting modifiers. Contribution B is a minimum comparability record that keeps nominal and measured composition, precursors, thermal history, environment, structure, transport definition, and replication in one account. Contribution C is a problem map in which each gap is written as an observed pattern, an evidence limitation, and a falsifiable next action.

The aim is interpretive discipline, not a performance leaderboard. We ask what the present collection permits a reader to associate, what it does not permit a reader to rank, and which missing fields prevent direct comparison. Figure 1 provides the navigation map; Table 1 provides the comparison rules. The scope is restricted to LLZO/LLZTO synthesis, phase stabilization, densification, process environment, and structure–transport evidence. Interface or cell observations appear only where their source record also reports processing-relevant LLZO information.

2 Scope, Evidence Method, and Comparison Rules

The evidence base is the repository’s exact audited library of 46 records covering 2007–2026 search intent, with foundational studies retained outside the emphasis on recent work. Metadata audit passed for all 46 entries, but evidentiary depth is uneven: 25 local records contain abstracts and 21 contain bibliographic metadata and titles only. We use an abstract-bearing record for a statement only when the statement is directly represented in that stored abstract. A title-only record may establish that the collection includes a paper framed around a named route, composition, or processing topic; it does not establish temperatures, property values, mechanisms, or comparative superiority. No sentence below denotes an exhaustive full-text review.

The collection includes studies that explicitly compare solid-state and sol–gel routes [14], relate hot-pressing conditions to density and separated transport readouts [4], compare crucible choices and air-exposure outcomes [37], and report phase, density, microstructure, and total conductivity together for a submicron Nb-containing garnet [40]. It also includes a cover-powder study whose abstract jointly reports lithium-rich firing conditions, density, total conductivity, activation energy, and grain-boundary composition [44]. These records motivate the fields in Table 1; they are examples of multi-field reporting, not a statement that every field was measured identically.

We classify comparisons in three tiers. *Directly comparable* requires the relevant minimum-record fields to match or be deliberately controlled. *Parallel qualitative evidence* applies when studies address the same question but differ in a known composition, process, or measurement field. *Insufficient fields* applies when a missing field could change interpretation. This rule is conservative by design: missing metadata in this collection are missing evidence, not proof that an author omitted an experiment or that a process variable had no effect.

3 Faceted Framework: Process, Modifiers, and Evidence

The framework is faceted, not a single MECE tree. Its first facet provides narrative order. P1 contains solid-state/precursor-control, solution-mediated, and coprecipitation/combustion powder routes. P2 separates Li-sublattice substitution, Zr-site or multi-element substitution, and stoichiometry/defect/phase questions. P3 separates pressureless schedules, pressure/field/ultrafast assistance, and low-temperature or additive-assisted densification. P4 contains phase/microstructure and transport readouts. These are primary reading anchors, not claims that a paper changes only one variable.

The second facet attaches modifiers across that chain: dopant identity and nominal site; starting lithium inventory and firing-stage lithium supply; crucible, cover powder, atmosphere, and exposure; and sintering additives or intergranular products. The third facet records evidence needed for comparison: composition and powder-route fields, thermal-process fields, structure/transport measurement fields, and replication or scale

Table 1: Minimum comparability record. “NR” means not reported in the checked source, never zero or absent.

Block	Required fields	Decision use
Composition and powder	Nominal and measured composition; dopant content; precursor identities; lithium excess; mixing; powder route	Separates formulation from processing and preserves an NR status for unobserved fields.
Thermal and environment	Calcination and sintering temperature/time; ramp; pressure or field; atmosphere; cover powder; crucible; additive	Prevents a route name from standing in for a complete thermal and chemical boundary condition.
Structure	Phase assemblage and method; density and basis; grains, boundaries, and porosity descriptors	Distinguishes achieved densification from an asserted densification mechanism.
Transport	Total, bulk, or grain-boundary conductivity; test temperature; electrodes; activation-energy convention	Blocks numerical comparison when the measured quantity or temperature is not aligned.
Reproducibility	Batch count; spatial sampling; specimen geometry; dispersion or uncertainty	Distinguishes one reported specimen from evidence about repeatability or uniformity.
Provenance	Title, stored abstract, local excerpt, or checked full text	Caps claim strength at the level actually inspected.

fields. This separation prevents an additive, for example, from being misdrawn as a serial stage between powder and pellet.

Three boundary tests make the framework operational. First, precursor control belongs to P1 when the controlled object is powder formation or mixing; composition and defect state belong to P2, while the same paper may retain both modifier and evidence tags. Second, “low-temperature densification” is a process objective in P3, whereas additive identity is an M4 modifier; a paper may legitimately carry both. Third, a modifier records what changed, while an evidence tag records what must be reported before the change can be compared.

Minimal phase and transport vocabulary is also necessary. The collection contains a computational abstract examining native defect populations as functions of synthesis conditions and explicitly challenging universal simple vacancy-compensation assumptions [33]. Other audited records are title-level anchors for gallium-associated cubic-phase stabilization [32], lithium stoichiometry in tetragonal LLZO [12], and LLZO synthesis/thermodynamics [1]. Because those three records are title-only locally, this review does not infer dopant occupancy, phase-transition values, or thermodynamic mechanisms from them. Likewise, total, bulk, and grain-boundary conductivity remain different observables; the word “conductivity” is not treated as a condition-complete comparison until type and test temperature are supplied.

The framework thus changes the reader’s default question. Instead of asking which route wins, it asks which stage was deliberately varied, which modifiers moved with it, which readout was used, and which minimum-record fields remain available. That question recurs in every route discussion below.

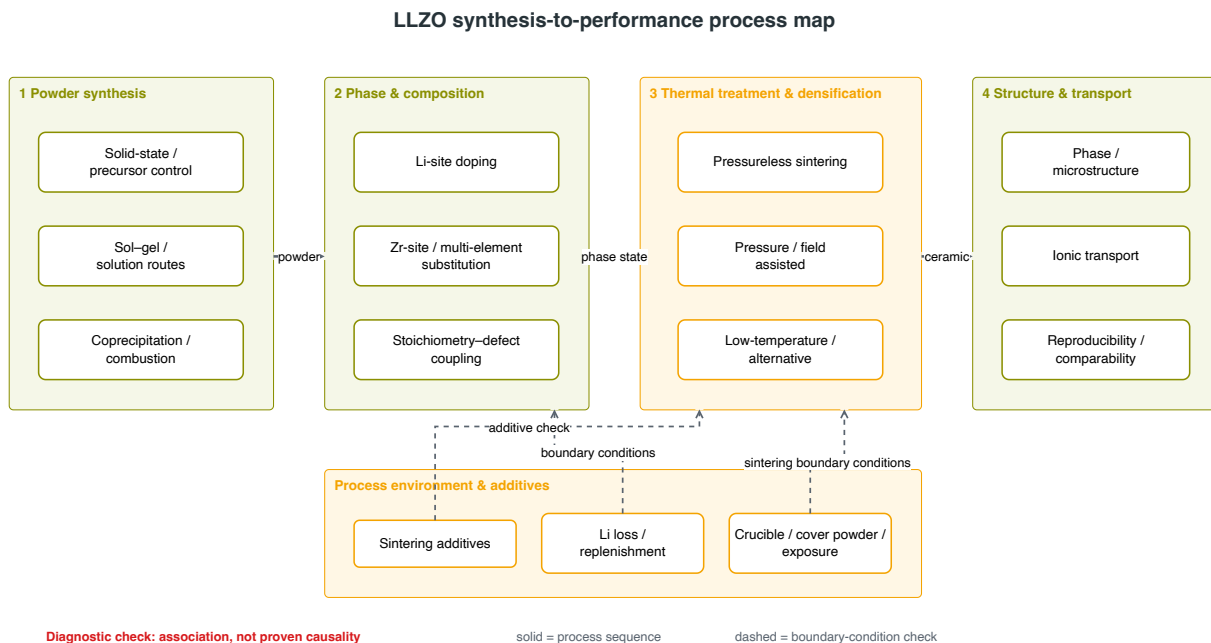


Figure 1: Faceted LLZO process map. Solid arrows encode group-level process order; dashed arrows encode boundary-condition checks involving lithium loss, crucible/exposure, and additives. The map identifies associations to inspect, not proven causal effects.

4 Powder-Synthesis Routes

4.1 Solid-state reaction and precursor control

In this framework, “solid state” names a powder-forming family, not a complete experiment. The audited set contains an early paper explicitly titled around synthesis of cubic LLZO for solid-state batteries [34]; because its local record is title-only, it anchors the existence of that route but not a particular recipe or outcome. A more information-rich abstract follows Al-containing LLZO through solid-state processing and reports that lithium inventory and powder surface-to-volume geometry are consequential within that study [8]. The contrast between those evidence levels is deliberate: the former locates a topic, while the latter supports an abstract-reported association.

Precursor identity can be varied without changing the route label. One stored abstract compares monoclinic and tetragonal zirconia starting powders under a reported common synthesis path and includes phase and transport readouts [23]. That record supports a controlled-within-study precursor comparison, but it does not make zirconia polymorph the only variable needed to compare against other studies. Likewise, in situ neutron diffraction is reported as tracking LLZO formation and identifying precursor-state and cooling-history features [25]. The latter is valuable because it treats phase formation as an observed sequence; any generalization beyond the studied composition and schedule would require its full methods and results.

Accordingly, the solid-state branch should be read through E1 fields: reagent identity and condition, lithium excess, mixing intensity, intermediate calcination, powder geometry, and provenance. If these fields differ or are unavailable, two “solid-state” specimens are parallel qualitative evidence, not direct competitors.

4.2 Solution-mediated and chelation routes

The collection contains title-level records for nitrogen-free sol-gel preparation of Al-substituted cubic LLZO [27] and for sol-gel synthesis coupled to lithium-ion-conductivity measurement [13]. Those titles establish solution-mediated coverage but do not license claims about lower temperature, finer powder, or superior transport. Such outcomes are conditional on the checked evidence.

Table 2: Evidence-matrix examples. The final column caps the inference at the locally checked provenance.

Primary tag	Intervention represented	Coupled readout	Licensed use here
P1.1	Zirconia precursor identity	Phase and transport	Within-study precursor comparison, abstract level [23]
P1.2	Solid-state versus sol–gel, with composition branch	Thermal, structural, electrochemical	Multi-factor comparative case, abstract level [14]
P1.3	Nonaqueous polymer combustion, undoped/Ta branch	Powder, sintering, transport	Route-to-readout case, abstract level [36]
P3.2	Hot-pressing temperature at fixed pressure	Density and transport partition	Within-study densification case, abstract level [4]
M3	Crucible choice and air exposure	Composition, density, transport	Boundary-condition comparison, abstract level [37]

Three abstract-bearing records permit more specific, still bounded statements. A solution-derived powder study reports subsequent pressure-assisted densification, showing that powder synthesis and pellet processing cannot be credited to one route label alone [28]. A comparative study places solid-state and sol–gel preparation, with and without Al, in one thermal, structural, and electrochemical analysis [14]. A PVP-chelated solution route reports predominantly cubic LLZO together with a secondary phase and an impedance-derived conductivity, so “phase obtained” should not be shortened to “phase pure” [2]. These records show why a solution route must be carried forward with its later densification and measurement record.

4.3 Coprecipitation, combustion, and rapid formation

A coprecipitation paper and a low-temperature powder-processing paper are title-only in the local corpus [7, 21]. A microwave-assisted solution-combustion paper is also title-only and is therefore used only to mark rapid powder formation for Ga-containing LLZO [3]. No temperature, phase fraction, particle-size advantage, or scale claim is assigned to those three records here.

By contrast, the stored abstract for a nonaqueous polymer-combustion study reports undoped and Ta-containing LLZO nanopowders and follows them into later sintering and transport characterization [36]. It therefore supplies an example of a route-to-readout chain rather than proof that combustion intrinsically outperforms a different route. The comparison lesson across P1.1–P1.3 is consistent: powder formation may alter mixing length scale, intermediate chemistry, or particle state, but a defensible route comparison must also preserve composition, thermal budget, green-body preparation, density, and transport definition. Where those common fields are absent, this review leaves the evidence parallel.

5 Composition and Phase Stabilization

Composition control is a cross-cutting facet because the nominal formula can change before, during, and after thermal processing. The collection includes title-level studies framed around Ga addition and cubic-phase stabilization [32], Ga substitution and grain coarsening [17], and lithium volatilization with phase change during Al-containing LLZO processing [19]. These titles identify three different control objects—substitution, microstructural development, and firing-stage inventory—but they do not establish site occupancy, phase fraction, or a causal chain in the present review.

The defect question is not reducible to a fixed vacancy count. The stored abstract of a first-principles study calculates native defect populations across synthesis conditions and reports that donor substitution does not yield the same simple lithium-vacancy compensation under all modeled conditions [33]. This is

theoretical evidence about a conditional defect landscape, not direct experimental proof for every processed pellet. It nevertheless justifies retaining nominal composition, measured composition where available, and synthesis environment as separate comparison fields.

Experimental abstracts reinforce the need for that separation. A precursor-homogeneity study reports changes in compositional and structural evolution and connects spatial mixing of reactants and dopant species to reproducible material characteristics within its experiments [22]. A Nb-containing submicron-powder study reports phase, relative density, microstructure, total conductivity, and activation energy under stated firing conditions [40]. An AlN-additive study on Ta-containing garnet jointly reports densification, transport, and electrochemical readouts and proposes reaction products as part of its interpretation [41]. In the present evidence hierarchy, the reported associations are usable; the detailed site chemistry or mediation mechanism would require full-text examination.

Zr-site and multi-element changes are similarly inseparable from processing. The local title for a dual-substitution study also names spark plasma sintering [5]. It would be invalid to attribute its outcome exclusively to composition or exclusively to densification without the full factorial context. The same restraint applies to title-only records on tetragonal-LLZO lithium stoichiometry [12] and lithium excess in cubic-LLZO synthesis [20]: they map M2 coverage but do not determine the sign or magnitude of an effect here.

For comparison, phase labels must therefore travel with at least four distinctions. First, nominal and measured compositions are not interchangeable. Second, a proposed substitution site is not treated as measured occupancy unless the checked source establishes it. Third, starting lithium excess is separated from lithium supplied or lost during firing. Fourth, a phase assemblage and its measurement method are retained rather than compressed into “cubic.” This record supports direct comparison only when the associated powder route, thermal history, environment, density, and conductivity definition are also compatible.

6 Thermal Treatment and Densification

6.1 Pressureless schedules

Pressureless firing is not a single baseline. The audited library contains a title-level study of pressure filtration followed by sintering for Al-containing LLZTO [16]; it marks shaping plus firing but provides no locally checked density or optical-property value. A stored abstract systematically varies lithium excess and dwell time in Ga-containing LLZO and reports that these parameters matter within its processing case [29]. Another abstract studies oxygen-assisted firing and reports void, density, conductivity, hardness, and cell-related readouts [31]. Taken together, these records show that schedule, lithium inventory, atmosphere, and green-body preparation must remain visible even when pressureless firing is the primary tag.

Uniformity adds a scale within the specimen. The stored abstract of a powder-covering strategy considers particle-size distribution, heating rate, support geometry, densification evolution, and spatially uniform readouts [6]. This is evidence that uniformity can be a measured target rather than an assumed consequence of average density. It does not establish that the reported strategy transfers unchanged to every composition or furnace.

6.2 Pressure, field, and ultrafast assistance

The title-level corpus contains papers explicitly framed around spark-plasma-sintering mechanisms [39], field-assisted densification of Al-substituted cubic LLZO [45], rapid ultra-high-temperature processing of Ga-containing LLZO [35], and ultrafast high-temperature sintering of LLZO [11]. These keys establish that pressure, field, and ultrafast branches are represented. They do not support energy-efficiency, mechanism, density, or conductivity rankings in this manuscript.

Two abstracts permit bounded comparisons. One compares solid-state sintering, SPS, and SPS followed by heat treatment using phase, morphology, and ionic-transport readouts [38]. Another varies hot-pressing temperature at a reported fixed pressure and reports relative density, grain-boundary resistance, and room-temperature total conductivity [4]. Each is informative within its own specimens. Across studies, however, starting powder, dopant content, pressure path, ramp, peak temperature, dwell, post-treatment, and measurement temperature differ or may be unavailable. We therefore do not rank “SPS,” “hot press,” “field assisted,” and “ultrafast” as universal technologies.

6.3 Low-temperature and additive-assisted densification

“Low temperature” is meaningful only relative to a comparator and material system stated in the same source. A cold-sintering abstract describes an LLZO-based composite, a transient liquid, microstructure, conductivity, and a cell demonstration [30]. Because this is a composite route, it is not silently normalized to monolithic LLZO firing. A SiO₂-addition abstract reports removal of surface carbonate, densification, grain-boundary transport, and cell readouts, while also advancing a mechanistic interpretation [42]. A CuO study reports a lower firing schedule for W-containing garnet together with microstructural, transport, wetting, and cell measurements [43]. A 2024 Li₅AlO₄-addition study reports a within-paper comparison involving firing temperature, relative density, conductivity, critical current, and cell operation [46]. These are associated outcomes from abstract-bearing records, not proof of one common densification pathway.

The evidence rule is therefore to record achieved density separately from the mechanism proposed to explain it. An additive may supply lithium, alter a transient phase, change boundary chemistry, or correlate with particle rearrangement; an abstract-level association cannot by itself discriminate those alternatives. A fair comparison must preserve parent powder, additive dose, green density, pressure, complete thermal history, final phase assemblage, density basis, and separated transport if available. Otherwise, “lower temperature” remains a study-specific descriptor rather than a league table.

7 Cross-Cutting Process Environment and Additives

The firing environment is part of the material history, not background metadata. In a stored abstract, otherwise related Al-containing pellets fired in alumina and platinum crucibles are compared using composition, relative density, ionic conductivity, and air-exposure readouts [37]. The record supports the statement that crucible choice and exposure were experimental variables in that study; it does not license a universal crucible prescription. A separate title-level record explicitly concerns alumina-crucible contamination associated with extra lithium carbonate [15], so it is retained as a topic anchor without importing a causal mechanism.

Lithium inventory must be split temporally. Starting excess is set in the batch, whereas lithium activity during firing can be influenced by atmosphere, cover material, geometry, and dwell. A title-only paper in the collection is framed around manipulation of a Li₂O atmosphere for dense LLZO [9]. An abstract-bearing cover-powder study reports LiOH-containing cover material together with density, total conductivity, activation energy, and grain-boundary composition [44]. Another abstract screens several lithium-containing, lower-cost cover-powder candidates for compensation of firing-stage lithium loss [10]. These records map related controls, but only a matched design could separate starting inventory from boundary supply.

Exposure continues after firing. A stored abstract presents lithium–proton exchange as a variable to be controlled for reproducible LLZO processing and performance [26]. That association makes humidity, exposure time, storage state, surface treatment, and delay before impedance measurement candidates for the minimum record. Their absence from a checked source is recorded as NR; it is not interpreted as zero exposure.

Additives create a second attribution problem. The title of a W-containing garnet paper explicitly links a sintering aid with microstructure and conductivity topics [18], but its local title-only provenance prevents

any detailed effect claim. Abstract-bearing studies associate SiO₂, CuO, AlN, or Li₅AlO₄ additions with combinations of densification, boundary or phase interpretations, transport, and cell readouts [41–43, 46]. Because additive identity, lithium supply, reaction products, density, and boundary transport may move together, no single abstract establishes which mediator is necessary or sufficient across compositions.

The practical comparison unit is consequently not “additive present.” It is the tuple {parent powder, nominal and measured additive content, mixing, green-body state, firing schedule and atmosphere, cover/crucible, phase assemblage, density basis, boundary characterization, conductivity type and temperature}. The uniform-densification abstract shows why geometry and spatial sampling also belong to that tuple [6]. With the tuple complete, a study may support a within-paper causal test; with only an additive name and an endpoint, it remains an association.

8 Structure–Transport Evidence and Minimum Comparability

Structure–transport reasoning often compresses a multistep chain into “denser means more conductive.” The audited abstracts do support study-specific covariation, but they also reveal why the shortcut is unsafe. A hot-pressing study reports density and the fraction of resistance associated with grain boundaries as pressing temperature changes [4]. A submicron Nb-containing study reports phase, relative density, microstructure, total conductivity, and activation energy together [40]. A crucible comparison reports composition, density, conductivity, and response to air exposure [37]. Each case involves more than a scalar density.

Relative density also does not uniquely encode pore location, grain size, boundary chemistry, surface state, or spatial uniformity. The abstract on uniform densification explicitly couples microscopic and macroscopic observations with heating and support conditions [6]. The solution/solid-state comparison attributes transport differences within its specimens to microstructural features while also varying route and Al inclusion [14]. We retain those as authors’ abstract-reported interpretations, not as a universal density–conductivity law.

Transport labels must remain explicit. Total conductivity includes the response resolved under the study’s equivalent-circuit treatment; bulk and grain-boundary terms are not interchangeable with it. Test temperature, electrode configuration, specimen geometry, frequency range, fitting choice, and activation-energy convention can change the reported observable. The cover-powder abstract is useful precisely because it identifies total conductivity and activation energy alongside density and grain-boundary composition [44]; it does not make those values directly comparable to a record that omits temperature or reports a different partition.

We therefore apply the minimum record in a fixed sequence:

1. Establish composition and provenance: nominal formula, measured chemistry when available, dopant and additive, and whether support comes from title, abstract, excerpt, or checked full text.
2. Establish material history: precursors, mixing and powder route, lithium excess, calcination, shaping, and every sintering field relevant to pressure, time, ramp, atmosphere, cover, and crucible.
3. Establish specimen state: phase method, density basis, grain and boundary descriptors, porosity or spatial sampling, and exposure history.
4. Establish transport meaning: total/bulk/grain-boundary quantity, temperature, electrodes, analysis convention, replicates, and uncertainty.

The output is categorical, not a score. If relevant fields align, two observations may be directly compared. If a known field differs, they remain parallel qualitative evidence that can motivate a controlled test. If a decisive field is NR, the comparison is insufficiently specified. This avoids two symmetrical errors: treating all nominally similar LLZO as identical, and treating every difference as evidence for the processing variable named in a title.

Contribution B is thus more than a reporting checklist. It is the decision layer attached to Contribution A's process map. Contribution C then uses the same fields to specify why an unresolved comparison remains unresolved. For example, a low-temperature additive case cannot discriminate density mediation from boundary-phase mediation unless density and boundary response are separately measured. A route comparison cannot isolate precursor homogeneity unless composition and thermal budget are matched. These are test-design consequences of the evidence map, not claims that the relevant experiments are absent from the field.

9 Six Evidence-Bounded Research Gaps

The gaps below are gaps in discrimination and comparability within the audited collection. They are not claims that an experiment has never been performed. Each begins with an abstract-level pattern, identifies the alternative explanations left coupled, and states an observation that could falsify the working explanation. This is Contribution C's problem map applied through Contribution B's minimum record.

9.1 Gap 1: precursor homogeneity versus powder route

Three stored abstracts provide complementary anchors: precursor homogeneity is deliberately examined in one study, solid-state and sol-gel preparation are compared in another, and a solid-state route is followed through lithium and powder-geometry controls in a third [8, 14, 22]. The present records do not establish a common-powder, matched-composition, matched-thermal-budget comparison with a shared replication protocol. Route chemistry, mixing scale, dopant distribution, and later firing may therefore explain the same apparent pattern.

A discriminating study would use one reagent lot and a predeclared factorial design in which mixing or powder route changes while nominal composition, dopant, green-body preparation, and final thermal schedule are held fixed. Independent batches would be assessed with the same phase, density, and separated transport protocol. The working hypothesis that greater precursor homogeneity reduces outcome dispersion would be falsified if the prespecified dispersion measure is not lower than the matched solid-state control. This action tests an explanation; it does not assume the conclusion.

9.2 Gap 2: starting lithium excess versus firing-stage lithium control

The abstracts for a solid-state route and a Ga-containing processing study report lithium inventory among their controlled variables, while the cover-powder abstract reports a lithium-rich firing boundary together with density and transport readouts [8, 29, 44]. Across those records, starting excess, dwell, specimen surface-to-volume ratio, and cover-powder supply are not isolated on one common composition.

A predeclared two-by-two design can separate nominal lithium excess (present/absent) from lithium-supplying cover material (present/absent), with common powder, crucible, geometry, and thermal history. The record should close a lithium balance as far as the selected analytical method permits and retain phase, density, and transport uncertainty. An interaction explanation would be weakened or falsified if its preregistered interaction term were indistinguishable from zero within the study's declared uncertainty. The gap is the absence of this discrimination in the checked collection, not an assertion of field-wide absence.

9.3 Gap 3: additive-enabled densification versus boundary-phase mediation

Abstracts for SiO₂, Li₅AlO₄, and AlN interventions each report multiple structural or electrochemical outcomes and include interpretations involving carbonate removal, lithium supply, or boundary products [41, 42, 46]. Because additive identity, dose, lithium activity, reaction products, final density, and boundary response change together, these abstracts cannot determine whether transport differences are mediated by density, boundary chemistry, or both.

A falsifiable next action is an additive-free control plus a dose series from one parent powder, with density matched where feasible and phase assemblage, boundary composition, and total/bulk/grain-boundary response measured before cell tests. A proposed boundary-phase mediation would be falsified if its predeclared boundary signature were absent or failed to covary with the separated grain-boundary response after density was accounted for. Null and contradictory outcomes would remain part of the report.

9.4 Gap 4: fair comparison of pressureless, pressure-assisted, and ultrafast firing

One abstract compares solid-state sintering, SPS, and SPS plus heat treatment, and another links a hot-pressing series to density and transport partition [4, 38]. Ultrafast high-temperature processing is represented locally only by a title-level record [11]. These provenance levels cannot support a cross-paper ranking, especially when starting powder, dopant, sample geometry, applied pressure, peak temperature, dwell, post-anneal, and conductivity temperature may differ.

A common characterized powder batch could be divided among pressureless, hot-press/SPS, and ultrafast protocols. Target density and specimen geometry would be declared in advance, energy/time exposure recorded, and one impedance protocol applied. The working equivalence hypothesis—that acceleration preserves transport at matched density—would be falsified if the accelerated route fell outside a prespecified equivalence margin relative to the pressureless control. This design compares routes on declared endpoints instead of method names.

9.5 Gap 5: exposure history as a reproducibility variable

Crucible and air-exposure outcomes are jointly reported in one stored abstract, lithium–proton exchange is presented as a processing-control variable in another, and precursor homogeneity is linked to reproducible material characteristics in a third [22, 26, 37]. The current collection does not expose humidity, duration, storage state, surface carbonate or proton-exchange indicator, and pre-measurement treatment as one shared reporting block. An apparent route or batch effect could therefore contain an uncontrolled exposure component.

Replicate pellets from one batch could be assigned to predeclared humidity–time histories, with a surface/compositional indicator measured before and after exposure and a common transport protocol repeated. The exposure hypothesis would be falsified if neither the selected surface-state metric nor transport changed systematically with cumulative exposure under the preregistered analysis. Reporting the null result would be as informative as reporting degradation.

9.6 Gap 6: an interlaboratory minimum record for density and conductivity

Abstract-bearing studies in the collection jointly expose particle-size and uniformity variables, phase/density/transport fields, and route/dopant comparisons [6, 14, 40]. They do not themselves constitute an interlaboratory round robin. Relative-density basis, spatial sampling, conductivity partition, test temperature, electrodes, replicate count, and uncertainty are not uniformly visible in the locally checked evidence.

A blind round robin on shared pellets and shared powder would permit each laboratory’s local protocol to be compared with a common minimum record. The preregistered analysis would partition dispersion associated with specimen, measurement, and reporting choices. The hypothesis that harmonized reporting and analysis explain a material fraction of between-laboratory dispersion would be falsified if the relevant variance component did not decrease. The immediate output would be a comparability result, not a claim that one laboratory or method is superior.

Across all six gaps, the common priority is controlled discrimination. The process map identifies where a variable enters; the minimum record preserves the conditions needed to interpret it; and the falsification clause states what evidence would change the working explanation. That combination avoids converting an evidence limitation into either a novelty claim or an experimentally actionable recipe.

10 Boxed Model-Assisted Precursor Diagnostic

General LLZO preparation literature provides context for why precursor identity and processing history matter [8, 24]; it does *not* validate the ranked sets below. The box reports a software-adaptor diagnostic from the bundled raw outputs, not a chemical conclusion.

MODEL-ASSISTED PRECURSOR DIAGNOSTIC — INFRASTRUCTURE DEMONSTRATION ONLY
CHEMICAL ROUTES UNVERIFIED — NOT FOR LAB USE

Target: $\text{Li}_7\text{La}_3\text{Zr}_2\text{O}_{12}$. The actual model-ranked sets are:

1. $\text{ZrO}_2 + \text{La}_2\text{O}_3 + \text{Li}_2\text{CO}_3$ (Top-1);
2. $\text{ZrO}_2 + \text{La}_2\text{O}_3 + \text{LiHO}$;
3. $\text{ZrO}_2 + \text{Li}_2\text{CO}_3 + \text{La}(\text{HO})_3$;
4. $\text{ZrO}_2 + \text{La}_2\text{O}_3 + \text{Li}_2\text{CO}_3 + \text{LiHO}$;
5. $\text{ZrO}_2 + \text{LiHO} + \text{La}(\text{HO})_3$.

These are model-ranked sets only. The strings LiHO and $\text{La}(\text{HO})_3$ are explicit nomenclature/entity-normalization warnings, not silently corrected chemical identities. Status is `model_output_verified=true`, `chemical_route_verified=false`, and `conditions=null`. A model score is not a reaction-success probability. No set above is a chemistry recommendation, no experimental conditions are supplied, and human evidence, entity, hazard, and waste review would be mandatory before any experimental plan could exist. **NOT FOR LAB USE.**

Raw provenance: `data/retro_llzo_top5.json` and `data/experiment_llzo_diagnostic.json`. Reviewer issue I4 remains open.

The diagnostic has one legitimate manuscript function: it makes the boundary between a verified software return and an evidence-qualified synthesis claim visible. It contributes neither chemical novelty nor an actionable route. The current entity warnings and absent conditions are failure signals to preserve, not fields to repair by inference.

11 Limitations

This is a standard-scale evidence-mapped review, not a comprehensive or systematic review. The 46-entry library passed bibliographic metadata audit, but only 25 local records include abstracts and 21 are title-only. Full-text inspection was selective, so a metadata PASS does not validate the context of every scientific claim. The manuscript consequently uses title-only entries for topic placement and labels abstract-reported associations at their actual provenance. It does not claim that the 46 records exhaust 2007–2026 LLZO work or that the small 2024–2026 subset represents current coverage.

The evidence matrix is intentionally condition-preserving rather than numerical. That choice prevents invented normalization but also means the review does not supply a performance meta-analysis. Missing fields are reported as unavailable in the checked source, not interpreted as experimental absence. The six gaps likewise diagnose what this collection cannot discriminate; full-text retrieval and a broader, reproducible search would be required before promoting them to field-wide absence claims.

The process map is a navigation asset derived from the adopted taxonomy. Its solid arrows denote process order and its dashed arrows denote boundary checks; neither encodes proven causality. The model-assisted box is more restricted still: its precursor sets are verified only as returned model output, its chemistry is unverified, its conditions are null, and the underlying reviewer issue I4 remains unresolved.

12 Conclusion

The central result is a disciplined way to read LLZO processing evidence. Powder synthesis, phase/composition control, thermal treatment/densification, and structure/transport form the primary chain, while composition, lithium inventory, environment, exposure, and additives cut across it. A separate evidence facet prevents those modifiers from being mistaken for stages or from disappearing behind route names.

The minimum comparability record converts that framework into a decision: results are directly comparable only when relevant composition, processing, structure, transport, and replication fields align; otherwise they remain parallel qualitative evidence or insufficiently specified. The six-gap map then points to three high-priority forms of discrimination: isolate starting lithium inventory from firing-boundary supply, separate density mediation from additive-related boundary chemistry, and harmonize density/transport records across specimens and laboratories.

These conclusions are calibrated to 46 audited records and selective abstract-level evidence. They do not rank a universally superior powder or densification route. Instead, Contributions A+B+C expose what was varied, what was measured, what traveled with the intervention, and what observation could overturn a working explanation. That is the level of synthesis the current evidence can support without turning missing fields, model outputs, or cross-paper incompatibility into chemical fact.

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